Journal of the Knowledge Economy (2024) 15:13182–13210
https://doi.org/10.1007/s13132-023-01623-z

Financing Health Systems in Developing Countries: the Role of Government Spending and Taxation

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Received: 2 January 2023 / Accepted: 13 November 2023 / Published online: 27 November 2023
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Abstract

This paper investigates the relationship between public health expenditure growth and macroeconomic determinants in 59 low-, low-middle-, and upper-middle-income countries from 2000 to 2019. The study is motivated by the need to understand how macroeconomic policies affect public health spending and the performance of health systems in these countries. To achieve this objective, the Arellano-Bover/Blundell-Bond system-GMM approach is applied on dynamic panel data by dividing the global panel into three sub-panels: full sample period, pre-global financial crisis period, and post-global financial crisis period. The results show that tax capacity and spending capacity exhibit positive impacts on public health expenditure in the full sample period and pre-global financial crisis period, respectively. Meanwhile, in the post-crisis period, public health expenditure is negatively influenced by the global financial crisis due to weak taxation capacity and high debt burden. We find that debt service burden has a positive impact on public health expenditure (% of GDP) in the full sample period and pre-global financial crisis period, respectively, while public health expenditure (% of GGE) is negatively affected by debt service burden in the post-crisis period. Furthermore, fiscal balance has a negative and significant impact on public health expenditure in all sample periods. Our analysis also shows that the rise of public health expenditures is significantly influenced by GDP per capita, population ageing, and development assistance for health. These findings have significant implications for policymakers seeking to improve health system financing and effectiveness in these countries, where governments should prioritize efforts to improve revenue and tax systems to allocate more funds towards health system financing performance.

Keywords Public health expenditure · Economic growth · Fiscal space · Tax revenue · Fiscal balance · Crisis

JEL Classification H5 · E62 · H2

Extended author information available on the last page of the article

Introduction

Delivering high-quality healthcare services that are affordable and equitably distributed has become a challenging objective for most low- and middle-income countries (LMICs). In a recent study, the United Nations (UN, 2019) noticed that at least half of the world's population still do not have full coverage of essential health services. What is even more concerning is that more than 100 million people are pushed into extreme poverty because of out-of-pocket payments (Dieleman et al., 2017; Kirton & Kickbusch, 2019). A recent report by the World Health Organization (WHO, 2019) noticed that about 925 million people spend more than 10% of their household income on healthcare, and 200 million people spend more than 25% of their income on health. These alarming statistics suggest that fighting poverty should ultimately go hand in hand with achieving universal health coverage[1] (UHC).

It is well established today that UHC helps achieve better health outcomes and allows a better individual financial protection by reducing out-of-pocket health expenditures (Marten et al., 2014; McIntyre et al., 2014; Pearson et al., 2016). Nevertheless, The WHO (2010) has already advised that it would be unlikely that countries can achieve UHC if they do not allow a minimum of 4–5% of their GDP as government health spending by 2030, although there is no clear framework that links this figure to the achievement of a particular health outcome.

In 2017, global health spending amounted to US$ 7.8 trillion (WHO, 2019) growing faster than the global economy. Public spending on health represents about 60% of global spending. Yet, its growth rate has been decelerating in recent years, from 4.9% on average between 2000 and 2010 to 3.4% in the 2010–2017 period. In per capita terms, public spending on health increased in all country income groups between 2000 and 2017, both as a share of public expenditure and as a share of total health spending, but with significant discrepancies across income groups. In low-income countries, the share of public health spending in gross domestic product (GDP) stagnated between 2000 and 2016, representing only 1.5% of GDP in 2016. In contrast, public health expenditures accounted for more than 3.5% of GDP in upper-middle-income countries and about 6% of GDP in high-income countries (Barroy et al., 2016).

While the upward trend in public health spending observed in high-income countries during the period 2000–2017 can be explained by the high prioritization of budget towards healthcare in these countries in the aftermath of the 2008 global financial crisis, stagnation of health spending as a share of government spending shows on the other side that in lower-middle-income countries budget prioritization for health plays a marginal role. The situation is worse in low-income countries, where economic growth and more public spending did not translate in higher allocations for health. In those countries, out-of-pocket expenditures and donor spending represent together 70% of total health spending (WHO, 2019). Ultimately, important external funding for health gives governments less incentives to prioritize health in their budgets.

[1] Universal health coverage mean all people receiving high-quality health services that meet their needs without exposing them to financial hardship in paying for the services.

Existing literature generally attributes the decreasing in resources allocated towards health system financing in poor countries to different macroeconomic factors like deprioritization of health sector, inefficiency in health expenditure, and low level of economic growth (Behera & Dash, 2019; Behera et al., 2022; Durairaj & Evans, 2010; Mathauer & Carrin, 2011; Novignon & Nonvignon, 2015; Tandon & Cashin, 2010). However, in this paper we believe, like others (Durairaj & Evans, 2010; Tandon & Cashin, 2010; Behera & Dash, 2019), that the pertinent factor for the slower growth of public expenditure on healthcare is due to less fiscal space in these economies.

Fiscal space for health has recently been recognized as an important issue that all countries should consider if they look forward making significant progress towards UHC (Barroy et al., 2016; Meheus & McIntyre, 2017; Barroy et al., 2018; Zeng et al., 2020; Akram & Rath, 2021). Firstly introduced by Heller in 2005 (Heller, 2005), fiscal space can be defined as the budgetary room that allows a government to allocate resources to the public sector without threatening the debt sustainability, that is to say without impacting government solvency given existing fiscal conditions and long-term requirements.

Based on Heller's definition, Tandon and Cashin (2010) identified five channels of fiscal space expansion for health including (i) conducive macroeconomic conditions, (ii) reprioritization of health within the government budget, (iii) an increase in health sector–specific resources (i.e., earmarked funds), (iv) health sector–specific grants and foreign aid, and (v) an increase in the efficiency of existing health expenditure.

Particularly, Tandon and Cachin (2010) argue that favorable macro-fiscal conditions could feasibly increase the share of the total budget allocated to health. Thus, during financial crises, the country's fiscal response may be sometimes so restrictive that it finally imposes the rationalization of public health spending. In the aftermath of the global financial crisis of 2008, health spending has stagnated across the leading industrialized nations, which has prompted policymakers to reconsider their fiscal space.

While extensive work has been conducted in the context of high-income countries (Maresso et al., 2015), there has been less attention paid to rigorously assessing fiscal space for health studies for LMICs. A challenging exercise in this paper would be to assess the extent to which public health expenditures of LMICs have been affected by fiscal pressures in response to the global financial crisis, although the literature generally considers that LMICs have been relatively less affected by the crisis than high-income countries (Leach-Kemon et al., 2012; Behera & Dash, 2019).

Using a panel data for the sample period of 2000 to 2019, this paper examines the dynamic effects of macroeconomic policies on public health expenditure growth for low-, low-middle-, and upper-middle-income countries. Our contribution to the empirical literature is fourfold. First, we employ an augmented fiscal space for health function to examine whether macro-fiscal factors impact the growth of public health expenditure for low-, low-middle-, and upper-middle-income groups. Second, we split the global panel into three sub-panels, that is, full sample, pre-global financial crisis period, and post-global financial crisis period. This classification would give clear insight regarding the macro-fiscal policies behavior towards health system financing in the surge of global financial crisis. Third, we use a panel dynamic system Generalized Method of Moment (GMM) estimates to assess the elasticity of

heath expenditure with respect to changes in macroeconomic factors. This technique has the soundness to capture the potential endogeneity problem via a set of instrumental variables (IV). Finally, for conducting the robustness check, we separate the sample into three subgroups, i.e., low-, low-middle-, and upper-middle-income groups according to the World Banks' standard income classification.

The remainder of this paper is structured as follows. The "Literature Review" section provides a review of the related literature. The "Data and Methodology" section describes the data and the econometric methodology. The "Empirical Results and Discussion" section discusses the empirical results. The "Conclusion and Policy Implications" section reports the conclusions and policy implications.

Literature Review

Public health expenditure is crucial for providing adequate healthcare services, particularly in LMICs. However, the financing of healthcare systems in LMICs is often challenging due to limited resources and competing priorities. This literature review explores the impact of macroeconomic determinants and economic downturns on public health expenditure. This literature review aims to contribute to a better understanding of the factors that influence public health expenditure in LMICs. It is divided into two subsections: the first focuses on the effects of macro-fiscal determinants on health expenditure, while the second examines the effects of economic downturns on health expenditure.

The Effects of Macro-fiscal Determinants on Public Health Expenditure

The literature on the impact of macro-fiscal determinants on public health expenditure is relatively rich and abundant. For instance, Sfakianakis et al. (2021) explore the responsiveness of public health expenditure to several macro-fiscal factors in 36 OECD countries and show that factors like GDP, fiscal capacity, tax revenues, and population ageing have a positive and significant impact on health financing during the period 2000 to 2017. Behera and Dash (2019) examine the impact of several macro-fiscal and sociodemographic factors on public health expenditure in 85 LMICs over the period 2000–2013 and found that tax revenues, GDP per capita, and population ageing are positively related to public health expenditure, while fiscal budget balance (surplus/deficit) and public debt services payments have a negative impact on public health expenditure. Younsi et al. (2016), using data from 167 countries from 1993 to 2013, find a positive effect of GDP, tax-financed health systems, social health insurance financing, and general government spending on public health expenditure. Reeves et al. (2015) investigate the tax revenues effects on health expenditure growth in lower-middle-income countries and show that tax revenues have a significant positive impact on public health investment. The GDP is found to be positively related to public health expenditure. This positive impact is mediated by greater tax revenues and increasing public expenditure. Castro and Camarillo (2014) provide evidence that countries having higher per capita GDP

also have larger possibility of generating higher tax revenues. Furthermore, Castro and Camarillo (2014) found a positive relationship between taxation capacity and health outcome (measured by life expectancy and infant mortality). Bajo-Rubio and Gomez (2015) report that increasing tax revenues and reducing unproductive expenditures are alternative strategies to boost health spending. However, it appears that an increase in tax revenues, especially through income tax and VAT rates, has a negative effect on GDP. Liang and Mirelman (2014) document that government debt and per capita GDP have a positive and statistically significant impact on public health expenditure. The positive sign on government debt implies that debt financing provides financial leverage to increase public expenditure on health. Conversely, Lu et al. (2010) show that per capita GDP and debt have negative and statistically insignificant relationship with government health expenditure as a share of GDP. Lora and Olivera (2007) find that higher debt ratios or higher debt services reduced health spending as well as social sector spending. In the meanwhile, a number of empirical studies have employed government spending as a share of GDP to predict the growth of public health spending (e.g., Hartwig & Sturm, 2014; Ke et al., 2011; Lu et al., 2010; Palangkaraya & Yong, 2009; Landon et al., 2006). For instance, Hartwig and Sturm (2014) provide evidence that the lagged values of government spending as % of GDP play a greater role in the growth of public health spending in the OECD countries. Ke et al. (2011) found a strong relationship between public health spending and spending capacity, measured by the government spending to GDP ratio, in low-income and middle-income countries. Particularly, prepayment services through tax-based systems positively affect government health spending in these countries. Lu et al. (2010) suggest that increases in government health expenditures are mainly due to the increasing spending capacity. They confirm that low spending capacity can lead to a longer time period to achieve better health outcomes. Landon et al. (2006) also found a positive and significant relationship between lagged health spending and current health spending.

The Effects of Economic Downturn on Public Health Expenditure

Calvo (2010) documents that most of developing economies have been severely involved by the global economic crisis by mainly increasing their fiscal deficits and debt services due to weak revenue generation, leading to cuts in social sector spending. Chubrik et al. (2011) examined the impact of global financial crisis on public health expenditure as a share of GDP in the Soviet Union Countries. They found that despite their poorer fiscal capacity, a number of countries have potted the public health expenditure split at slightest at pre-crisis level. Kirigia et al. (2011) showed that government health expenditure in most African countries has been negatively affected by the global economic crisis, because the massive reduction in their budgetary allocation towards health sector. But the crisis rebellion created new health policy measures such as allocating more revenue to health development, increasing domestic funding for the health sector, and mobilizing more revenue from external funds. Enkelejda (2012) investigated the impact of global economic crisis on the healthcare system in the case of Albania and provided evidence that government

spending on health is reduced by the reduction of domestic revenue and health budget cut from the total government budget. Simou and Koutsogeorgou (2014) analyzed the effects of the economic crisis on healthcare in Greece from 2009 to 2013. They found that between 2009 and 2011, the annual public health expenditure has been reduced by 20% points, resulting in a subsequent reduction in the health workforce, salaries and pensions, and the supply of medical goods and services.

Data and Methodology

Data

To examine the impact of macroeconomic policies on public health expenditure growth, this study uses annual panel data of 11 low-income, 22 lower-middle-income, and 26 upper-middle-income countries from 2000 to 2019. A complete list of the selected countries is provided in Table 4 (see Appendix).

Following preceding studies (e.g., Barros, 1998; Behera & Dash, 2019; Getzen, 1992; Jaunky & Khadaroo, 2008; Ke et al., 2011; Landon et al., 2006; Lu et al., 2010; Younsi et al., 2016), we consider our dependent variables as public health expenditure (% of GDP) and public health expenditure (% of GGE). Following Heller (2006), Durairaj and Evans (2010), Tandon and Cashin (2010), Ke et al. (2011), Bajo-Rubio and Gómez-Plana (2015), Barroy et al. (2016), Younsi et al. (2016), Meheus and McIntyre (2017), and Behera and Dash (2019), we have chosen to focus entirely on the fiscal space which includes tax revenue (% of GDP), total debt service (% of GDP), and fiscal balance (% of GDP). Hence, the effect of such macro-fiscal factors on public health expenditure would exhibit the pattern of health system financing in low-, low-middle-, and upper-middle-income countries and would be able to determine the retroactive determinants of revenue, which could have an additional impact on health spending growth. In line with previous studies (Baltagi & Moscone, 2010; Barro, 2013; Braendle & Colombier, 2016; Carrion-i-Silvestere, 2005; Hartwig & Sturm, 2014; Jaunky & Khadaroo, 2008; Santiago et al., 2013; Younsi et al., 2016; Zhike & Huiming, 2014), we consider GDP per capita, general government expenditure (% of GDP) (Schieber & Maeda, 1999; Lu et al., 2010, Younsi et al., 2016), and development assistance for health (Baker, 2010; Farag et al., 2009; Ke et al., 2011; Mishra & Newhouse, 2009; Younsi et al., 2016), which is essential factors for low-, low-middle-, and upper-middle-income countries for financing their public healthcare system. Moreover, consistent with earlier studies (Bloom et al., 2015; Getzen, 1992; Ke et al., 2011; Meijer et al., 2013; Palangkaraya & Yong, 2009; Younsi et al., 2016; Zweifel et al., 1999), we include the population ageing (% of the total population),[2] which exhibits the demand for health services and thereby increases the allocation of the health budget in developing countries.

All data are from the World Bank's World Development Indicators (WDI) database (World Bank, 2020). Definition of the variables and data sources are shown in Table 5 (see Appendix). Table 1 provides the summary statistics of the considered variables.

[2] The ageing population is usually considered a burden to society and policies should be adapted to offer financial support for healthcare to this special category of people.

Overall, we observe that, on average, GDP per capita accounts for $465.450 in the low-income countries for the period 2000 to 2019, while it accounts for $1656.512 and $5191.463 in lower-middle- and upper-middle-income countries, respectively. However, during the pre-crisis period, all income groups experienced a decrease in GDP per capita but a significant increase over the post-crisis period. Public health expenditure reached an average of 1.216% of GDP in the low-income countries, with a slight increase (+0.051%) during the pre-crisis period, while it accounts for 2.039 and 3.274%, respectively, in the lower-middle- and upper-middle-income countries with a slight increase in average over the post-crisis period. Indeed, the analysis reveals that variation in health expenditure significantly depends on the country's economic growth level. However, some countries spend more than 15% of GDP on health, while others spend less than 5.5% on health. Regarding public health expenditure as a share of GGE, we observe that low-income countries reached an average of 6.470%, with an increase (+0.475%) during the pre-crisis period and a decrease (−0.701%) during the post-crisis period, while lower-middle- and upper-middle-income countries reached an average of 7.868 and 10.997%, respectively, with a moderate decline (−0.093 and−0.446%, respectively) during the pre-crisis period and a moderate increase (+0.199 and+0.661%, respectively) during the post-crisis period. General government expenditure accounted an average for 11.746% of GDP in the low-income countries, with a slight decrease (−0.106%) over the pre-crisis period and a moderate increase (+0.195%) over the post-crisis period, while it accounted for 13.724 and 15.620% of GDP, respectively, in the lower-middle- and upper-middle-income countries. It is observed that in both the lower-middle- and upper-middle-income countries, general government expenditure suffered a slight drop (−0.868 and−0.079%, respectively) over the pre-crisis and a slight increase (+1.104 and+0.116%, respectively) over the post-crisis period. However, we find that upper-middle-income countries spend more than low- and lower-middle-income countries. Tax revenue in low-income countries stood at an average of 10.123% of GDP with a moderate decline (−0.909%) over the pre-crisis period and a significant increase (+1.447%) over the post-crisis period. However, the tax-to-GDP ratios in the lower-middle- and upper-middle-income countries were higher than in the lower-income countries, at 14.574 and 18.176% of GDP, respectively. Ultimately, we notice that the descriptive statistics of debt service burden, fiscal balance (surplus/deficit), and development assistance for health also vary significantly over the different sub-panels.

Moreover, the correlation matrix of variables presented in Tables 6, 7, 8, and 9 (see Appendix) shows that fiscal balance is negatively and significantly correlated with public health expenditure as a share of GDP in the full sample and all income groups, while there is a negative but statistically insignificant relationship between public health expenditure as a share of GGE and fiscal balance in all income groups. Tax revenue and general government expenditure are strongly correlated with public health expenditure as a share of GDP in all income groups. There is also a positive relationship between public health expenditure as a share of GDP and debt services in the full sample and lower-middle-income countries. GDP per capita is positively and significantly correlated with public health expenditure as a share of GDP and public health expenditure as a share of GGE only in the full sample and lower-middle-income countries.

Table 1 Summary statistics of low-, lower-middle-, and upper-middle-income countries: pre- and post-global financial crisis period

Variables		Low-income countries			Lower-middle-income countries			Upper-middle-income countries		
		Full sample (2000–2019)	Pre–crisis (2000–2008)	Post–crisis (2011–2019)	Full sample (2000–2019)	Pre–crisis (2000–2008)	Post–crisis (2011–2019)	Full sample (2000–2019)	Pre–crisis (2000–2008)	Post–crisis (2011–2019)
GDP per capita (in current USD)	Mean	465.450	341.499	656.175	1656.512	1168.122	2436.090	5191.463	3832.755	7277.374
	Std. Dev	209.600	139.943	176.500	994.342	689.145	1004.705	2578.175	1873.574	2342.748
	Min	111.927	111.927	354.480	258.471	258.471	861.758	474.217	474.217	3757.559
	Max	1162.905	820.151	1162.905	4438.688	4010.857	4438.688	13,890.630	10,854.172	13,890.630
PHE (% of GDP)	Mean	1.216	1.267	1.145	2.039	1.915	2.215	3.274	3.085	3.528
	Std. Dev	0.630	0.695	0.531	1.218	1.082	1.363	1.253	1.219	1.242
	Min	0.062	0.062	0.343	0.331	0.331	0.434	1.148	1.148	1.397
	Max	3.145	3.145	2.363	5.833	5.630	5.800	6.763	6.763	6.664
PHE (% of GGE)	Mean	6.470	6.945	5.769	7.868	7.775	8.067	10.997	10.551	11.658
	Std. Dev	3.288	3.546	2.706	4.002	3.836	4.280	4.613	4.472	4.666
	Min	1.212	1.212	1.370	1.701	1.701	2.762	2.738	2.738	3.615
	Max	18.321	18.321	15.610	20.833	20.575	20.833	33.462	32.734	33.124
TAX (% of GDP)	Mean	10.123	9.214	11.570	14.574	13.971	15.545	18.176	18.136	18.387
	Std. Dev	3.343	2.873	3.857	5.298	5.084	5.365	6.086	6.110	6.239
	Min	0.780	0.780	1.778	5.611	5.611	8.498	4.994	4.994	8.629
	Max	18.954	15.369	18.954	39.258	38.532	35.237	45.253	45.253	40.681
GGE (% of GDP)	Mean	11.746	11.640	11.941	13.724	12.856	14.828	15.620	15.541	15.736
	Std. Dev	4.667	4.906	4.414	6.880	4.899	9.310	3.982	4.191	3.697
	Min	2.058	2.057	5.398	0.952	0.951	5.039	7.195	7.195	9.237
	Max	25.794	25.794	23.192	74.270	35.826	74.270	26.242	25.689	26.242
DEBT (% of GNI)	Mean	5.930	6.902	4.754	13.344	13.611	13.156	17.553	17.972	17.298
	Std. Dev	4.578	4.906	3.667	8.693	7.993	9.793	14.432	14.357	14.706
	Min	0.231	0.231	0.717	0.494	0.759	0.494	0.382	0.722	0.382
	Max	25.729	25.729	17.242	55.104	43.541	55.104	88.050	87.004	88.049

Table 1 (continued)

Variables		Low-income countries			Lower-middle-income countries			Upper-middle-income countries		
		Full sample (2000–2019)	Pre–crisis (2000–2008)	Post–crisis (2011–2019)	Full sample (2000–2019)	Pre–crisis (2000–2008)	Post–crisis (2011–2019)	Full sample (2000–2019)	Pre–crisis (2000–2008)	Post–crisis (2011–2019)
FBAL (% of GDP)	Mean	−11.725	−14.682	−6.539	−55.310	−59.188	−39.502	−41.288	−41.165	−23.269
	Std. Dev	33.358	38.509	18.991	84.605	96.509	53.721	124.806	143.417	73.284
	Min	−212.320	−212.320	−112.394	−469.844	−469.844	−276.369	−889.777	−889.777	−185.691
	Max	44.203	32.063	26.185	118.532	118.532	29.862	525.382	525.382	296.758
POP ageing (in %)	Mean	3.064	3.038	3.115	5.573	5.445	5.780	7.201	6.875	7.737
	Std. Dev	0.637	0.526	0.787	3.366	3.330	3.418	3.882	3.743	4.099
	Min	2.174	2.189	2.174	2.465	2.502	2.470	2.280	2.358	2.280
	Max	5.476	4.660	5.476	16.051	16.051	15.886	20.076	17.732	20.076
DAH (in million USD)	Mean	23.501	19.907	28.135	8.309	7.680	9.050	3.023	3.655	2.217
	Std. Dev	13.388	11.364	13.686	9.354	7.840	11.098	20.600	27.329	3.111
	Min	3.250	3.250	4.020	0.130	0.13	0.140	0.000	0.000	0.000
	Max	67.000	53.250	65.310	52.240	38.450	52.240	469.000	469.000	14.520

Authors' calculations based on data from WDI

Theoretical Framework and Model Construction

The construction of the econometric model is inspired from the theoretical framework of Tandon and Cashin (2010). The authors depict the fiscal space for health sector using a typical government's intertemporal budget constraint which can be described as the following algebraic equation:

$$G_t + r_t B_{t-1} = T_t + B_t + A_t + O_t \tag{1}$$

Equation (1) states that the utilization of budgetary sources (aggregate expenditure) like government non-interest expenditure G_t and non-discretionary debt interest payments $r_t B_{t-1}$ should be equal to the sources of budgetary resources (aggregate revenue) such as tax revenues from different sources T_t, total government borrowings B_t, grants from national and international sources A_t, and other sources of funds from non-tax revenues O_t. After the aggregate revenue generation from different budget sources in a given year, the fiscal space for health might be contacted through the proportion k_t of public health expenditures regarding aggregate expenditures G_t. Equation (1) can thus be expressed in the reduced form as follows:

$$PHE_t = k_t G_t \tag{2}$$

Equation (2) states that if G_t grows on account of an increase in the overall factors of resource mobilization (i.e., tax revenue, non-tax revenue, grants, and borrowings), then the *PHE* would rise by a fixed proportion k_t of the government's budgetary commitment on health sector remains unchanged. Hence, the proportion change in *PHE* as regards *G* shows the financial commitment of the government to the health sector in a given year. That is to say, the *PHE* trends over time are in effect related to the conducive government's macro-fiscal policies which mobilize the resources in the regards health sector through sustained economic growth, greater revenue mobilization, lower debt level, and lower fiscal deficit (Heller, 2006; Durairaj & Evans, 2010; Tandon & Cachin, 2010).

Thus, our objective is to access the effect of resource mobilization using increasing growth and deriving health prioritization. The prioritization of health expenditure is negatively influenced by a huge fiscal gap, which results in a large debt supply and debt services payments coming out in low- and low-middle-income countries (Barroy et al., 2016). However, continued and expanded country's fiscal deficit and debt services payment reduces the government's ability to fulfil its primary liability for evolving socioeconomic infrastructure (Baldacci et al., 2012; Lora & Olivera, 2007).

Referring to the above theoretical background, we estimate an augmented fiscal space for health function similar to that of Tandon and Cashin (2010), Younsi et al. (2016), and Behera and Dash (2019), where public health expenditure is a function of macro-fiscal policies and a set of control variables:

$$PHE_{it} = f(GDP_{it}, TAX_{it}, GGE_{it}, DEBT_{it}, FBAL_{it}, POP_{it}, DAH_{it}) \tag{3}$$

where *PHE* is public health expenditure, *GDP* is gross domestic product per capita, *TAX* is tax revenue to GDP ratio, *GGE* is general government expenditure to GDP ratio,

DEBT is total debt services to GNI ratio, *FBAL* is a fiscal balance to GDP ratio (+surplus/−deficit), *POP* is population ageing, and *DAH* is development assistance for health.

Estimation Strategy

To account for potential endogeneity problems in all regressors and capture lagged influences, one should apply linear dynamic panel data model-Generalized Method of Moments (GMM) which is the most commonly used method to eliminate fixed effects in the case of small time series and large cross-sectional units. We therefore use the system-GMM procedures[3] (Arellano & Bover, 1995; Blundell & Bond, 1998) to analyze the dynamic effects of macroeconomic policies on the growth of public health expenditure. The dynamic equation is as follows:

$$Y_{it} = \alpha_i + \gamma Y_{i,t-1} + \beta X_{it}\prime + \varepsilon_{it} \tag{4}$$

where $i = 1, 2,\ldots, N$ is the number of selected countries and $t = 1, 2,\ldots, T$ is the time. Y_{it} denotes the dependent variable, X_{it} represents the vector of exogenous variables, α, β, and γ are the coefficients to be estimated, and ε_{it} is the error term to capture the residual variation. However, in Eq. (4), X_{it} may contain accurate exogenous variables as well as some types of weekly exogenous variables or endogenous variables. For instance, X_{it} is assumed to be exogenous when $E\left(X_{it}\varepsilon_{it}\right) = 0$ for all i and t, and it is assumed to be endogenous when $E\left(X_{it}\varepsilon_{it}\right) \neq 0$ for all i and t. On the other side, if X_{it} have reverse relationship with ε_{it},X_{it} can be viewed as endogenous.

In our empirical approach, we have assumed that public health expenditure as a share of GDP, public health expenditure as a share of GGE, and tax revenue are endogenous variables because current taxation capacity may depend on its lagged tax collection (Behera & Dash, 2019; Hartwig & Sturm, 2014; Landon et al., 2006). On the other hand, macroeconomic factors like total debt service, fiscal balance, and income per capita are considered to be weekly exogenous. Hence, the panel dynamic GMM models are constructed as follows:

$$\begin{aligned} \ln PHE/GDP_{it} = \alpha_i + \gamma \ln PHE/GDP_{i,t-1} + \beta_1 \ln GDP_{it} + \beta_2 \ln TAX_{it} + \beta_3 \\ \ln GGE_{it} + \beta_4 \ln DEBT_{it} + \beta_5 FBAL_{it} + \beta_6 \ln POP_{it} + \beta_7 \ln DAH_{it} + \varepsilon_{it} \end{aligned} \tag{5}$$

$$\begin{aligned} \ln PHE/GGE_{it} = \alpha_i + \gamma \ln PHE/GGE_{i,t-1} + \beta_1 \ln GDP_{it} + \beta_2 \ln TAX_{it} + \beta_3 \\ \ln GGE_{it} + \beta_4 \ln DEBT_{it} + \beta_5 FBAL_{it} + \beta_6 \ln POP_{it} + \beta_7 \ln DAH_{it} + \varepsilon_{it} \end{aligned} \tag{6}$$

All variables are taken in logarithms, except for the fiscal balance. ln denotes the natural logarithm, *PHE/GDP* shows the public health expenditure as a share of GDP, *PHE/GGE* shows the public health expenditure as a share of general government expenditure, *GDP* shows the gross domestic product per capita, *TAX* is

[3] Arellano and Bover (1995) and Blundell and Bond (1998) have proposed the use of a system-GMM in dynamic panel data models that combines differenced equations with the level equation. The instruments for the level equation are lagged differences of the variables, which are valid when these differences are uncorrelated with the individual effects.

the tax revenue as a share of GDP, *GGE* is the general government expenditure as a share of GDP, *DEBT* is the total debt service as a share of GNI (Gross National Income), *FBAL* implies cash surplus/deficit as a share of GDP, *POP* represents the population ageing as a share of the total population, *DAH* denotes the development assistance for health while α, γ,β_1–β_7 are the coefficients to be estimated, and ε_{it} is the error term.

Estimation of Eqs. (5) and (6) with OLS regression techniques might lead to several econometric problems. For instance, including a lagged dependent variable in the regression could lead to an autocorrelation problem. The time-invariant fixed effects in the error term might be correlated with the explanatory variables. More importantly, OLS results may be biased because of potential endogeneity among the explanatory variables. The endogeneity problem can be tackled by adopting instrumental variables (IV) estimation. However, with weak instruments, the fixed-effects IV estimators are likely to be biased similar to OLS estimators. Instead of IV estimation, one could apply the system-GMM estimators (Arellano & Bover, 1995; Blundell & Bond, 1998) instead of difference-GMM (Arellano & Bond, 1991). As argued by Blundell and Bond (1998), the lagged dependent variable in difference-GMM estimation may accentuate the weak instruments bias. Alternatively, the system-GMM estimation technique gives more efficient estimators.

In our model specification, we apply the two-step robust system-GMM procedure to find efficient estimators. Indeed, the two-step estimator is more efficient than the default one-step estimator, as it has a smaller asymptotic variance and more powerful statistical tests. Furthermore, one-step GMM estimators use weight matrices that are independent of estimated parameters, whereas the efficient two-step GMM estimator yields a finite-sample corrected two-step covariance matrix. The soundness of the instruments is assessed by two diagnostic tests. First, the Hansen (1982) J-test is used for over-identification restrictions. The null hypothesis states that the instruments are uncorrelated with the error term. The second diagnostic test includes the second-order autoregressive AR (2) test that detects autocorrelation in the level series. Accordingly, the results presented in Table 2 confirm that all our regression specifications are robust and free from over-identification restrictions and second-order autocorrelation (AR2) (see columns 1–6).

Empirical Results and Discussion

System-GMM Estimation Results: Public Health Expenditure (% of GDP)

The findings of our empirical analysis, as shown in Table 2, highlight the impact of macroeconomic factors on PHE as a share of GDP. Specifically, our results reveal that tax revenue has a positive and statistically significant impact on PHE growth in both the full sample and the pre-crisis period, respectively. It shows a 1% increase in tax revenue contributes to an increase in PHE as a share of GDP by 0.085 (full sample) and 0.112% (pre-crisis period), respectively (see columns 1 and 2). This result matches the findings of Lora and Olivera (2007), Bajo-Rubio

Table 2 Impact of macroeconomic determinants on PHE: system-GMM estimation results

	Dependent: ln(PHE/GDP)			Dependant: ln(PHE/GGE)		
	(1)	(2)	(3)	(4)	(5)	(6)
Variables	Full sample	Pre-crisis	Post-crisis	Full sample	Pre-crisis	Post-crisis
$\ln(PHE/GDP)_{t-1}$	0.482***	0.550***	0.385***			
	(0.0483)	(0.0671)	(0.0690)			
$\ln(PHE/GGE)_{t-1}$				0.565***	0.672***	0.438**
				(0.0489)	(0.0674)	(0.0651)
ln(GDP)	−0.064***	−0.062***	−0.070**	0.098**	0.142**	0.085
	(0.0255)	(0.0529)	(0.0573)	(0.0248)	(0.0415)	(0.0682)
ln(TAX)	0.085***	0.112***	−0.097*	−0.122***	−0.242***	0.225***
	(0.0148)	(0.0394)	(0.0309)	(0.0410)	(0.0534)	(0.0669)
ln(GGE)	0.535***	0.258***	0.352***	0.450***	0.528***	0.095*
	(0.0462)	(0.0498)	(0.0375)	(0.0397)	(0.0729)	(0.0510)
ln(DEBT)	0.003*	0.011***	−0.003***	0.054***	0.053**	−0.075***
	(0.0041)	(0.0184)	(0.0082)	(0.0126)	(0.0207)	(0.0126)
FBAL	−0.003***	−0.002***	−0.009***	−0.002	−0.002	0.005**
	(0.0001)	(0.0001)	(0.0002)	(0.0001)	(0.0001)	(0.0001)
ln(POP)	0.112**	0.198**	0.445**	−0.083**	−0.053**	−0.091**
	(0.0804)	(0.0403)	(0.0542)	(0.0618)	(0.0458)	(0.0592)
ln(DAH)	0.003*	0.004*	0.004*	0.065**	0.011**	0.021**
	(0.0252)	(0.0332)	(0.0253)	(0.0277)	(0.0276)	(0.0291)
Constant	−2.366***	−1.577***	−1.883***	−0.236***	−0.216	0.338
	(0.219)	(0.152)	(0.184)	(0.0787)	(0.142)	(0.251)
AR(2) test (*p-value*)	0.492	0.225	0.402	0.514	0.258	0.359
Hansen test (*p-value*)	0.662	0.443	0.591	0.693	0.369	0.576
Observations	1029	536	353	1029	536	353
No. of country	59	55	57	59	55	57

Hansen J-test for over-identified restriction for the instrument used in the model; the values reported for AR(2) are the *p*-values for second-order autocorrelated disturbances in the first differences equations; corrected standard errors are in parentheses

*significant at 10%; **significant at 5%; ***significant at 1%

Author's estimations

and Gómez-Plana (2015), Reeves et al. (2015), and Behera and Dash (2019), suggesting that tax revenue is strongly associated with greater investment in PHE, resulting in higher levels of social security and more access to healthcare services. However, in the post-crisis period, this relationship is negative and statistically significant. Similarly, GGE shows a significant positive association with PHE as a share of GDP in all model specifications, meaning that a 1% increase in GGE leads to an increase in PHE as a share of GDP by 0.535 (full sample), 0.358 (pre-crisis period), and 0.352% (post-crisis period), respectively (see columns 1–3). Debt service burden shows significantly reduce PHE in the post-crisis

period, revealing that a 1% increase in the debt-to-GDP ratio leads to a 0.003% decrease in PHE as a share of GDP. This result is consistent with the findings of Lora and Olivera (2007) and Landon et al. (2006), suggesting that high debt services reduce PHE as well as social sector expenditure. However, in the full sample and pre-crisis period, this relationship is positive and statistically significant.

Moreover, we observe that fiscal deficit situation harms the PHE as a share of GDP in all model specifications. This implies that a 1% increase in the level of fiscal deficit leads to a 0.003 (full sample), 0.002 (pre-crisis), and 0.009% (post-crisis period) reduction in PHE as a share of GDP. This confirms that the increasing fiscal deficit in a given financial year might decrease the allocation health budget in the long term. This result corroborates that of Lora and Olivera (2007) and Behera and Dash (2019), who suggested that certain fiscal policy interventions in terms of alternative revenue generation would be the potential source for health financing in the long term. Yet, this result is inconsistent with some prior studies that prove that fiscal balance has not any direct effect on PHE rather fiscal deficit may indirectly affect the allocation health budget through the lower level of economic growth (Baldacci et al., 2008; Landon et al., 2006; Liu et al., 2010; Murthy and Okunade, 2000). Furthermore, our analysis indicates that per capita GDP exhibits a statistically significant negative association with PHE as a share of GDP in all model specifications. This implies that PHE is not procyclical, but rather counter-cyclical. The results suggest that higher growth rates in LMICs do not necessarily lead to increased budgetary allocation towards the health sector. Furthermore, since both variables are entered into the regression as natural logs, the estimated coefficients can be interpreted as an elasticity. This negative relationship indicates that governments in LMICs perceive public health as an inferior good. This result highlights the need for policymakers to prioritize PHE in times of economic growth and expansion to ensure adequate financing for the health sector. This result stands in contrast with the findings of several earlier studies. Arze del Granado et al. (2013), Chen et al. (2013), Hartwig and Sturm (2014), Velenyi and Smitz (2014), and Liang and Tussing (2019) have all reported that government health expenditures (GHE) tend to be procyclical. On the other hand, Afonso and Jalles (2013) and Behera and Dash (2019) found that GHE shows no clear pattern relative to GDP, making it acyclical. Similarly, Chubrik et al. (2011) and Doytch et al. (2012) have found that GHE is procyclical in middle-income countries, but acyclical in low-income countries.

The negative and statistically significant correlation between per capita GDP and PHE found in this study highlights an important issue related to government priorities towards healthcare. Despite the higher economic growth rates in LMICs, the negative relationship between per capita GDP and PHE suggests that governments do not prioritize spending on health as a key area for development. This is concerning, especially in light of the current global health challenges and the increasing demand for accessible and affordable healthcare services. Governments need to allocate adequate resources to fund PHE and ensure that the health sector receives the attention it deserves.

Besides, the empirical analysis reveals that lagged PHE as a share of GDP has a positive and statistically significant effect on current PHE. This implies that a 1% increase in PHE a share of GDP in the last year leads to less than 1% change

in the current year allocation of the health budget. This provides evidence for path dependence in PHE, whereby current spending is influenced by past levels. Thus, future health status depends on the current year's PHE which in turn depends on the previous financial year's budget allocation. This empirical result is like the findings of Landon et al. (2006), Liu et al. (2010), Barro (2013), Younsi et al. (2016), and Behera and Dash (2019), suggesting that the past allocation of PHE is handled as an "endogenous variable" and thus determines the present allocation of PHE. Furthermore, the impact of lagged PHE as a share of GGE appears to be stronger compared to the lagged PHE as a share of GDP. These findings demonstrate that past investments in public health have a crucial role in determining the current level of spending, underscoring the importance of sustained commitments to the sector.

The empirical evidence presented in Table 2 reveals that population ageing drives increases in PHE as a share of GDP. This implies a 1% increase in the share of the aged population leads to a rise in PHE by 0.112 (full sample), 0.198 (pre-crisis period), and 0.445% (post-crisis period), respectively. This finding suggests that as populations age, governments allocate more resources towards PHE. This result is consistent with the findings Palangkaraya and Yong (2009), Ke et al. (2011), Meijer et al. (2013), Bloom et al. (2015), Behera and Dash (2019), and Younsi and Bechtini (2020), who suggested that demographic changes explain the rising of healthcare services demand and healthcare costs, which lead to an increase in the allocation health budget. However, this finding contradicts those of Zweifel et al. (1999), Sen (2005), and Hartwig and Sturm (2014), who found that the ageing population is not associated with higher public expenditure on healthcare. Furthermore, our analysis shows that DAH has a positive and statistically significant impact on PHE as a share of GDP. This result implies that external resources play a critical role in supplementing domestic PHE. It highlights the importance of international aid and cooperation in achieving UHC in LMICs.

System-GMM Estimation Results: Public Health Expenditure (% of GGE)

The estimation results reported in Table 2 show significant effects on PHE based on various factors. Firstly, the lagged effect of PHE as a share of GGE has a positive impact on the reprioritization of health expenditure in all model specifications, indicating the importance of path dependency in determining current health expenditure levels. In other words, past levels of spending on public health can influence current spending decisions, and this effect appears to be stronger when measured relative to government expenditure rather than GDP.

Furthermore, debt service burden has a positive and significant impact on PHE as a share of GGE in the full sample and pre-crisis period. This implies that a 1% increase in debt service burden leads to a 0.054 (full sample) and 0.053% (pre-crisis period) increase in PHE as a share of GGE. This result indicates that the servicing of debt does not necessarily crowd out spending on healthcare. This finding may be of interest to policymakers who are concerned about the trade-offs between servicing debt and providing social services. However, a significant negative relationship is found over the post-crisis period. Fiscal balance has a significant positive effect on

PHE as a share of GGE in the post-crisis period, while it has a marginally negative and insignificant effect in the full sample and pre-crisis period (see columns 4–6). This finding suggests that governments in LMICs with higher fiscal balances tend to allocate a larger share of their government expenditure towards public health. This may be due to the increased availability of funds for the government to invest in public health. However, the effect of fiscal balance on PHE as a share of GGE is not consistent over time and may be influenced by other factors. Therefore, it is important for policymakers to consider the current fiscal balance of their country when prioritizing PHE, especially in times of economic crisis when government resources may be more limited. Our analysis also reveals that tax revenue has a significant and negative impact on PHE as a share of GGE in the full sample and pre-crisis period (see columns 4 and 5). This relationship is found to be positive and significant in the post-crisis period. This implies that a 1% incremental change in tax revenue contributes to a 0.225% change in PHE as a share of GGE.

Moreover, the empirical evidence reveals that GGE has a positive and statistically significant impact on PHE as a share of GGE in all sample periods. We find that a 1% increase in GGE leads to an increase in PHE as a share of GGE by 0.450 (full sample), 0.528 (pre-crisis period), and 0.095% (post-crisis period), respectively. Per capita GDP also has a positive and significant impact on PHE as a share of GGE in the full sample and pre-crisis period, but this relationship becomes slightly positive and insignificant in the post-crisis period. This result indicates that economic growth is a driver of health spending, but other factors, such as austerity measures, have constrained the government's ability to increase health spending in the face of economic growth.

Besides, population ageing has a negative impact on PHE as a share of GGE in all sample periods. This result implies that a 1% increase in the share of the ageing population leads to a 0.083 (full sample), 0.053 (pre-crisis period), and 0.091% (post-crisis period) reduction in PHE as a share of GGE, respectively. This finding suggests that an ageing population may be putting pressure on the government's health budget. In contrast, DAH exerts a positive and significant impact on PHE as a share of GGE in all sample periods, which indicates that external sources of funding can play an important role in supporting health spending in LMICs. However, reliance on such funding may not be sustainable in the long term, and policymakers may need to consider how to increase domestic resources for health. Similar results are presented in previous literature (Baker, 2010; Farag et al., 2009; Kirigia et al., 2011; Mishra & Newhouse, 2009; Younsi et al., 2016).

Overall, these results highlight the importance of using PHE as a share of GGE as a measure of spending prioritization, as it more accurately reflects the government's budget capacity and the value of healthcare in LMICs. The positive effect of total debt service on PHE also suggests that debt service burden does not negatively affect the reprioritization of health expenditure.

Robustness Check

The robustness check results presented in Table 3 suggest that the relationship between PHE and fiscal variables varies across income groups. In lower-middle- and upper-middle-income countries, we find that PHE is significantly and positively associated with taxation capacity, but this relationship becomes marginally positive and insignificant in low-income countries. Specifically, a 1% increase in tax revenue as a share of GDP is associated with a substantial increase in PHE of 0.185 (low-middle-income) and 0.122% (upper-middle-income), respectively. This implies that low-, lower-middle-, and upper-middle-income countries depend on taxation capacity to finance their health expenditures, and thus all these countries need to increase their revenue mobilization towards health financing through improving taxes from income and wealth and widening the tax base.[4] Additionally, our analysis shows that PHE is significantly and positively related to spending capacity in lower-middle- and upper-middle-income countries. This means that an increase of 1% in GGE as a share of GDP leads to a significant increase in PHE of 0.208 (low-middle-income) and 0.442% (upper-middle-income), respectively. This expresses that PHE in lower-middle- and upper-middle-income countries largely depends on spending capacity.

Moreover, we find that PHE is more sensitive to the debt services in lower-middle- and upper-middle-income countries, with a 1% increase in total debt service leading to a reduction in PHE of 0.032 (low-middle-income) and 0.024% (upper-middle-income), respectively. However, the negative effect of total debt service on PHE growth highlights the importance of managing debt burden to ensure sustainable financing of health expenditures in these countries. In contrast, in lower-income countries, we find that PHE is slightly less sensitive to the debt burden, as a 1% increase in total debt service reduces PHE by 0.015%. This suggests that low-income countries have sufficient fiscal space to manage both debt repayment and health expenditures. Our analysis also suggests that PHE is more sensitive to the fiscal balance (-deficit) in all income countries. We find that fiscal deficit has a significant negative effect on PHE. It implies that a 1% increase in fiscal deficit leads to a 0.002 (low-income), 0.005 (low-middle-income), and 0.004% (upper-middle-income) reduction in PHE. This result suggests that these countries rely more on fiscal capacity to finance their health expenditures. Additionally, this finding suggests that the decrease in public revenues is the main reason behind the higher government deficits associated with the financial crisis. Hence, importance needs to be given to fiscal consolidation measures such as an increase in public revenue and

[4] Referring to the literature, the low- lower-middle, and upper-middle-income countries might be generated revenue mobilization towards health financing through, for example, various direct and indirect taxation solutions: (i) government-owned enterprises or assets (e.g., revenue comes from natural resources such as oil, natural gas and other liquids, and earmarked tax on tobacco and alcohol (McCoy, 2009; Jha et al., 2012; Bajo-Rubio & Gómez-Plana, 2015; WHO, 2016); (ii) compulsory earmarked tax revenues such as taxes on personal income and corporate income, profits, and taxes on sales and custom duties (Bitran, 2012; Reeves et al. 2015; Reich et al., 2016); and (iii) non-compulsory taxes levied on value-added-tax receipts from the business agreement, currency transactions, tobacco purchase, mobile phone minutes purchase and airline tickets purchase (Rabi, 2014; Barroy et al. 2016).

Table 3 System-GMM estimation results for the three income groups: robustness check

Variables	Dependent: ln(PHE/GDP)			Dependant: ln(PHE/GGE)		
	Low-income panel	Low-middle-income panel	Upper-middle-income panel	Low-income panel	Low-middle-income panel	Upper-middle-income panel
$\ln(PHE/GDP)_{t-1}$	0.562***	0.751***	0.559***			
	(0.0904)	(0.0646)	(0.0762)			
$\ln(PHE/GGE)_{t-1}$				0.548***	0.762***	0.724***
				(0.0824)	(0.0648)	(0.0626)
ln(GDP)	−0.005*	−0.015*	0.073**	0.132	0.014	0.044**
	(0.0118)	(0.0284)	(0.0169)	(0.1020)	(0.0252)	(0.0198)
ln(TAX)	0.053	0.185**	0.122***	0.208	0.102	0.075
	(0.0457)	(0.0890)	(0.0658)	(0.175)	(0.095)	(0.0784)
ln(GGE)	0.155	0.208**	0.442***	0.285	0.092	0.237**
	(0.0897)	(0.0985)	(0.0967)	(0.213)	(0.0747)	(0.0958)
ln(DEBT)	−0.015	−0.032***	−0.024**	0.012	0.038*	−0.020
	(0.0442)	(0.0245)	(0.0133)	(0.0397)	(0.0232)	(0.0162)
FBAL	−0.002**	−0.005***	−0.004***	−0.006	−0.002	−0.005
	(0.0010)	(0.0010)	(0.0010)	(0.0009)	(0.0001)	(0.0001)
ln(POP)	−0.015	0.085*	0.092*	0.067**	−0.016*	−0.086*
	(0.0502)	(0.0412)	(0.0436)	(0.0514)	(0.0674)	(0.0658)
ln(DAH)	0.021*	0.009*	0.024	0.011**	0.012**	0.005*
	(0.0980)	(0.0382)	(0.0112)	(0.0112)	(0.0392)	(0.0145)
Constant	−1.809	−1.587***	−1.770***	−0.930	−0.348	−0.658**
	(1.723)	(0.549)	(0.323)	(1.438)	(0.228)	(0.270)
AR(2) test (*p-value*)	0.269	0.460	0.547	0.466	0.293	0.435
Hansen test (*p-value*)	0.584	0.398	0.849	0.578	0.560	0.694
Observations	178	367	483	178	367	483

Table 3 (continued)

Variables	Dependent: ln(PHE/GDP)			Dependant: ln(PHE/GGE)		
	Low-income panel	Low-middle-income panel	Upper-middle-income panel	Low-income panel	Low-middle-income panel	Upper-middle-income panel
No. of country	11	22	26	11	22	26

Hansen J-test for over-identified restriction for the instrument used in the model; the values reported for AR(2) are the *p*-values for second-order autocorrelated disturbances in the first differences equations; corrected standard errors are in parentheses

*significant at 10%; **significant at 5%; ***significant at 1%

Author's estimations.

unproductive spending cuts when the deficit is large and even more if the economy follows a financial crisis (Baldacci et al., 2012; Reinhart & Rogoff, 2009).

Moreover, the results indicate that per capita GDP has a negative and significant effect on PHE growth in low- and low-middle-income groups. It shows that a 1% decrease in per capita GDP results in a decrease of 0.005 (low-income) and 0.015% (low-middle-income) in PHE growth, respectively. In upper-middle-income countries, however, this relationship is found to be statistically significant and positively related to PHE growth. Besides, we find a positive and significant relationship between per capita GDP and PHE as a share of GGE in upper-middle-income countries, with a 1% increase in per capita GDP leading to an increase in PHE as a share of GGE of 0.044%. This result matches the findings of Farag et al. (2009) and Younsi et al. (2016). In low- and lower-middle-income countries, GDP per capita exerts a positive but statistically insignificant effect on PHE as a share of GGE. The results also reveal that DAH has a positive and significant relationship with PHE as a share of GGE in all income groups. The ageing population gives an adverse impression to influence PHE, particularly in low-middle- and upper-middle-income countries, indicating that governments should respond to the demand for healthcare services by investing more funds to the health sector.

Conclusion and Policy Implications

Using panel data from 59 low-, low-middle-, and upper-middle-income countries from 2000 to 2019, this study investigates the determinants of public health expenditure, with a focus on the impact of taxation capacity, fiscal capacity, debt burden, and development assistance for health. The Arellano-Bover/Blundell-Bond system-GMM approach is applied on dynamic panel data by splitting the global panel into three sub-panels: full sample period, pre-global financial crisis period, and post-global financial crisis period. The empirical findings reveal that lagged public health expenditure, debt service burden, per capita GDP, fiscal balance, and development assistance for health are important determinants of public health expenditure. Furthermore, the robustness check provides further evidence that taxation capacity, spending capacity, and debt burden play a critical role in determining public health expenditure in different income groups.

This study contributes to the literature on public health expenditure by providing new empirical evidence on the determinants of public health financing in LMICs, with a focus on the critical role of fiscal policy (i.e., government spending and taxation). The findings emphasize the need for policymakers to prioritize public health expenditure and consider factors such as taxation capacity, fiscal capacity, debt burden, and development assistance for health to enhance public health spending and improve health outcomes in LMICs.

The results of this study have significant implications for policymakers. Firstly, policymakers should prioritize the allocation of resources to public health expenditure to improve health outcomes in LMICs. Secondly, they should consider increasing the fiscal capacity of their countries through measures such as increasing tax revenue to sustain public health spending in the long term. Thirdly, policymakers should ensure that

they manage their debt burden effectively, considering that high levels of debt service can reduce public health spending. Fourthly, policymakers should consider seeking development assistance for health to supplement domestic resources and enhance public health spending. Finally, the study highlights the importance of understanding the varying impacts of fiscal capacity, debt burden, and taxation capacity on public health expenditure in different income groups, which should inform targeted policy interventions to promote public health spending in each group.

To provide further valuable insights into this line of literature, future research could benefit from this study by including more countries, particularly developed countries, and other macroeconomic measures that influence public health financing. Additionally, comparative studies, i.e., developing versus developed countries before and after the global financial crisis, might be interesting.

Appendix

Table 4 List of countries

Lower-income countries: Afghanistan, Burkina Faso, Cambodia, Congo Dem. Rep., Ethiopia, Madagascar, Mali, Nepal, Rwanda, Togo, Uganda
Lower-middle-income countries: Armenia, Bangladesh, Bhutan, Cote d'Ivoire, Egypt Arab Rep., El Salvador, Georgia, Ghana, Guatemala, Honduras, India, Indonesia, Kenya, Kyrgyz Republic, Moldova, Morocco, Nicaragua, Pakistan, Philippines, Sri Lanka, Ukraine, Zambia
Upper-middle-income countries: Algeria, Angola, Belarus, Belize, Bosnia and Herzegovina, Botswana, Brazil, Bulgaria, Colombia, Costa Rica, Dominican Republic, Iran Islamic Rep., Jamaica, Jordan, Kazakhstan, Lebanon, Mongolia, Paraguay, Peru, Romania, Serbia, South Africa, St. Lucia, Thailand, Tunisia, Turkey

Table 5 Definition of the variables and data sources

Variables	Definitions	Source
PHE/GDP	Public health expenditure (% of GDP)	World Bank, WDI (2020)
PHE/GGE	Public health expenditure (% of General government expenditure)	World Bank, WDI (2020)
GDP	Gross domestic product per capita (GDP in current US dollars)	World Bank, WDI (2020)
TAX	Tax revenue (% of GDP)	World Bank, WDI (2020)
GGE	General government expenditure (% of GDP)	World Bank, WDI (2020)
DEBT	Total debt services (% of GNI: Gross National Income)	World Bank, WDI (2020)
FBAL	Fiscal balance (+surplus/−deficit (% of GDP))	World Bank, WDI (2020)
POP	Population ages 65 and above (% of the total population)	World Bank, WDI (2020)
DAH	Development assistance for health (in millions US dollars)	World Bank, WDI (2020)

Table 6 Pairwise correlation of variables (full sample)

	FBAL	PHE/GDP	PHE/GGE	GGE	TAX	DEBT	POP	GDP	DAH
FBAL	1.000								
PHE/GDP	−0.115***	1.000							
PHE/GGE	−0.0487	0.476***	1.000						
GGE	0.0078	0.462***	0.224***	1.000					
TAX	0.110***	0.449***	0.196***	0.348***	1.000				
DEBT	−0.202***	0.231***	0.147***	0.058*	0.081**	1.000			
POP	−0.0644**	0.356***	0.216***	0.221***	0.230***	0.339***	1.000		
GDP	−0.0581*	0.347***	0.350***	0.152***	0.323***	0.322***	0.345***	1.000	
DAH	0.048	−0.147***	−0.107***	0.039	−0.125***	−0.208***	−0.278***	−0.318***	1.000

*significant at 10%; **significant at 5%; ***significant at 1%

Author's estimations

Table 7 Pairwise correlation of variables (low-income countries)

	FBAL	PHE/GDP	PHE/GGE	GGE	TAX	DEBT	POP	GDP	DAH
FBAL	1.000								
PHE/GDP	−0.016*	1.000							
PHE/GGE	−0.081	0.465***	1.000						
GGE	0.319***	0.298***	0.035	1.000					
TAX	0.144*	0.271***	0.045	0.315***	1.000				
DEBT	0.227***	0.033	0.022	0.102	−0.110	1.000			
POP	−0.126*	−0.142**	−0.025	−0.342***	−0.035	0.322***	1.000		
GDP	−0.001	−0.031	−0.130*	0.101	0.391***	−0.316***	0.134*	1.000	
DAH	0.126*	0.280***	0.237***	0.298***	0.130*	0.002	−0.188**	0.214***	1.000

*significant at 10%; **significant at 5%; ***significant at 1%

Author's estimations

Table 8 Pairwise correlation of variables (lower-middle-income countries)

	FBAL	PHE/GDP	PHE/GGE	GGE	TAX	DEBT	POP	GDP	DAH
FBAL	1.000								
PHE/GDP	−0.019**	1.000							
PHE/GGE	−0.024	0.456***	1.000						
GGE	−0.005	0.414***	0.178***	1.000					
TAX	0.221***	0.305***	0.145**	0.323***	1.000				
DEBT	−0.217***	0.094*	0.162***	−0.023	−0.014	1.000			
POP	−0.084*	0.280***	0.102**	0.158***	0.187***	0.345***	1.000		
GDP	0.204	0.131**	0.207***	0.038	0.182***	0.212***	0.305***	1.000	
DAH	0.126***	0.151***	−0.007	0.264***	0.232***	−0.351***	−0.307***	−0.343***	1.000

*significant at 10%; **significant at 5%; ***significant at 1%

Author's estimations

Table 9 Pairwise correlation of variables (upper-middle-income countries)

	FBAL	PHE/GDP	PHE/GGE	GGE	TAX	DEBT	POP	GDP	DAH
FBAL	1.000								
PHE/GDP	−0.154***	1.000							
PHE/GGE	−0.035	0.442***	1.000						
GGE	−0.026	0.420***	0.141***	1.000					
TAX	0.140***	0.241***	−0.055	0.312***	1.000				
DEBT	−0.180***	0.017	−0.054	−0.107**	−0.145***	1.000			
POP	−0.006	0.342***	0.058	0.165***	0.027	0.157***	1.000		
GDP	−0.075*	0.042	0.068	−0.115**	−0.073*	0.129***	0.175***	1.000	
DAH	−0.033	0.058	0.032	0.108**	0.045	−0.016	−0.089**	−0.079*	1.000

*significant at 10%; **significant at 5%; ***significant at 1%

Author's estimations

Acknowledgements The authors gratefully acknowledge the valuable suggestions received from the Editor-in-Chief and two anonymous referees on the earlier draft of this paper, which substantially improved the paper.

Author Contribution MY contributed to the study design. MY analyzed the data, and MB verified the results. MY drafted the first version of the manuscript, and MB edited the manuscript. Both authors read and approved the final manuscript.

Data Availability Data will be made available on request.

Declarations

Ethics Approval and Consent to Participate Not applicable.

Consent for Publication Not applicable.

Competing Interests The authors declare no competing interests.

Statement of Human and Animal Rights Not applicable.

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